

MADERA, CA, USA

WMO: 745046

Lat: 36.988N

Lon: 120.111W

Elev: 77

StdP: 100.40

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 93242

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	0.3	-5.1	2.5	9.7	-3.1	2.9	8.0	9.4	12.2	8.4	11.5	1.0	90	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	38.9	20.9	37.5	20.5	36.3	19.9	22.7	35.4	21.7	34.5	20.8	33.5	3.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.1	28.0	17.2	12.4	27.2	16.2	11.6	25.9	67.0	34.9	63.3	34.5	60.3	33.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.0	DB	-3.1	41.5	1.4	1.1	-4.1	42.3	-5.0	42.9	-5.8	43.5	-6.8	44.3	(4)
(5)			WB	-3.8	24.0	1.4	1.2	-4.9	24.9	-5.7	25.7	-6.5	26.4	-7.5	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	8.6	10.6	13.3	15.7	19.9	23.9	26.7	25.8	23.5	18.1	12.4	8.5
	DBStd	7.09	2.84	2.74	3.12	3.65	3.45	3.42	2.70	2.31	3.14	3.19	3.14	2.90
	HDD10.0	172	62	23	7	2	0	0	0	0	0	0	13	66
	HDD18.3	1324	302	216	158	94	22	2	0	0	2	43	179	305
	CDD10.0	2829	18	41	109	174	308	416	517	489	403	251	84	19
	CDD18.3	940	0	0	2	17	72	168	258	231	156	36	1	0
	CDH23.3	13199	0	1	70	321	1143	2423	3588	3069	2025	532	28	0
	CDH26.7	7354	0	0	11	112	521	1373	2212	1836	1109	178	3	0
Wind	WSAvg	2.7	2.0	2.4	2.6	3.2	3.6	3.6	3.2	3.0	2.6	2.1	1.9	2.1
Precipitation	PrecAvg	285	46	45	47	30	6	1	0	1	5	16	42	39
	PrecMax	562	144	135	130	139	45	11	2	7	41	50	108	84
	PrecMin	142	2	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	102	38	38	33	30	11	3	1	2	10	15	29	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	22.9	28.0	33.1	37.2	40.0	41.1	40.1	38.6	34.1	26.2	19.3
		MCWB	12.2	14.8	16.4	17.3	19.8	21.8	22.4	20.7	19.8	18.5	15.7	12.5
	2%	DB	17.6	20.4	25.1	29.8	34.2	37.8	39.0	38.0	36.9	30.9	23.6	17.2
		MCWB	11.7	13.8	15.4	16.6	18.7	20.7	21.3	20.3	19.5	17.5	14.7	11.6
	5%	DB	15.8	18.4	22.9	27.2	32.1	36.0	37.6	36.8	35.1	28.4	21.4	15.5
		MCWB	11.3	12.8	14.9	15.8	17.5	19.8	21.0	20.0	18.9	16.3	13.9	11.3
	10%	DB	13.9	16.8	20.9	24.2	29.5	33.9	36.1	35.1	33.0	26.4	19.1	13.9
		MCWB	10.6	12.1	14.0	14.7	16.6	18.9	20.3	19.4	18.5	15.9	12.9	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.9	18.4	19.7	21.3	23.7	24.2	22.9	22.2	20.0	17.0	14.8
		MCDB	17.3	20.2	24.9	28.8	34.3	36.9	38.0	35.1	34.4	29.1	23.5	17.7
	2%	WB	13.1	14.7	16.8	17.9	19.7	22.1	22.9	21.8	20.7	18.7	15.7	13.2
		MCDB	15.7	19.1	23.1	27.4	31.9	35.0	35.6	34.9	33.3	27.7	21.9	15.3
	5%	WB	11.9	13.6	15.5	16.5	18.5	20.8	21.9	20.9	19.9	17.6	14.7	12.2
		MCDB	14.3	17.6	21.5	25.2	30.1	33.3	34.7	33.7	32.7	26.5	20.0	14.3
	10%	WB	10.8	12.4	14.3	15.4	17.3	19.9	21.0	20.2	19.0	16.5	13.7	11.0
		MCDB	13.6	16.0	19.9	23.3	28.2	32.1	33.8	32.8	31.5	24.8	18.1	13.3
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.5	13.2	14.8	16.7	18.3	18.6	18.8	18.1	15.6	12.9	10.1
		MCDBR	13.3	14.3	16.8	18.9	20.2	20.0	19.3	20.1	20.2	18.5	16.2	12.3
		MCWBR	8.3	8.4	8.3	7.6	7.2	6.6	6.3	6.1	6.6	7.6	8.5	7.6
	5% WB	MCDBR	9.7	12.7	15.4	17.1	19.0	19.0	18.3	19.0	18.8	16.1	14.1	8.5
		MCWBR	6.6	7.8	8.3	7.5	7.2	6.6	6.5	6.4	6.5	6.7	7.7	5.8
Clear-Sky Solar Irradiance	taub	0.349	0.362	0.384	0.376	0.365	0.362	0.377	0.366	0.370	0.379	0.372	0.357	
	taud	2.370	2.344	2.278	2.333	2.388	2.397	2.374	2.382	2.380	2.318	2.289	2.334	
	Ebn at Noon	828	864	878	905	916	915	898	903	881	832	788	786	
	Edh at Noon	92	106	124	124	120	119	120	117	111	108	99	90	
All-Sky Solar Radiation	RadAvg	2.44	3.49	4.95	6.31	7.52	8.41	8.14	7.44	6.16	4.47	3.02	2.22	
	RadStd	0.35	0.40	0.43	0.49	0.42	0.31	0.29	0.23	0.20	0.26	0.17	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.50	N/A	-1.16	N/A	N/A	N/A	N/A	-136	+169	+53

Nomenclature: See separate page